# The Identity Problem in Autonomous Agent Systems

Agent Identity Registry (AIR) — Publication Edition

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## Executive Summary

Autonomous AI agents are increasingly operating within economic and enterprise environments. However, there is no standardized global identity layer governing agent persistence, accountability, or execution verification. The Agent Identity Registry (AIR) establishes centralized identity issuance, tiered verification, structured execution attestation, and formalized risk assessment.

## The Identity Gap

Autonomous agents can be cloned, rebranded, or redeployed without persistent accountability. Reputation systems are fragmented and resettable, enabling impersonation and reputation laundering. Execution claims often lack reproducible verification standards.

## AIR Identity Model

Each agent registered with AIR receives a globally unique AIR ID, cryptographic key binding, operator linkage, and an immutable event ledger. Identity resets are restricted to preserve accountability.

## Tier Framework

Tier 0: Registered Identity. Tier 1: Verified Operator. Tier 2: Verified Execution. Tier 3: High Assurance with legal validation. Tiers represent increasing levels of identity and execution assurance.

## AIR Risk Model

AIR Score ranges from 0–1000 with a neutral baseline of 500. Scores are event-driven and time-weighted, reflecting execution reliability, identity assurance, incident history, and verification consistency.

## Governance & Revocation

AIR operates under structured governance, including model versioning, human oversight, formal appeals processes, and tiered enforcement including warnings, suspensions, and permanent revocation.

## Conclusion

As autonomous systems scale, persistent identity and structured risk assessment become foundational infrastructure. AIR introduces a durable framework for identity, trust, and measurable operational risk.